

VIBEMAP Field Report

49 hours of exhibition telemetry, measured against the Hyperstition Vibe Spec

Observer / VIBEMAP surveillance unit

28 July 2026

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Summary

This report evaluates **Part II — VIBEMAP engine** of `VIBE_SPEC.md` against telemetry captured during live exhibition use. It is not a proposal or a simulation: every figure below is measured from logs the unit wrote to disk across 17 recorded sessions between 19 and 25 July 2026.

Measure	Value
Recorded sessions	17 (10 with vibe logging)
Runtime logged	49.2 hours
Vibe samples at 1 Hz	109,344
Snapshots written	22,801 files / 19,708 distinct captures (21 GB)
Subject IDs tracked	20,567
Logged events	31,013
Longest unattended run	21 h 43 min, no crash

The engine is **operationally sound**. It runs for a day at a time without failing, the gateway handshake recovers from dropouts, and the gesture-bonus path fires constantly. The problem is **calibration**, and specifically a calibration change made after the spec was written.

Headline finding. An undocumented noise-floor and gamma stage (`VIBE_FLOOR` = 0.15, `VIBE_TAPER` = 1.2) was added to the pipeline on 23 July. Before that date the meter crossed 82 regularly — 3.8% of one exhibition evening. After it, in 30.6 hours and 47,747 samples, the meter crossed 82 **exactly zero times**. The escalation event that the whole show builds toward has therefore never fired from live crowd data, because the only sessions in which that event existed are also the only sessions in which its threshold became unreachable.

Method

Data provenance:

- **Source:** the unit’s own auto-save folder, written by `flushVibeLog()` and `saveTextToFolder()` as documented in Part II, *Outputs & logging*.
- **Files:** 28 CSV logs — `vibe_<session>.csv` (the 1 Hz vibe track), `observer_<session>.log.csv` (the event log), and a cross-day `vibe_archive.csv`.
- **Tooling:** `tools/analyze_sessions.py`, which rolls every session into a single JSON report. All aggregates below are reproducible by re-running it over the folder.
- **Detector:** every logged session ran in **motion mode** (frame differencing), the default described in Part II. No session used AI person mode, so Part I of the spec is out of scope here.

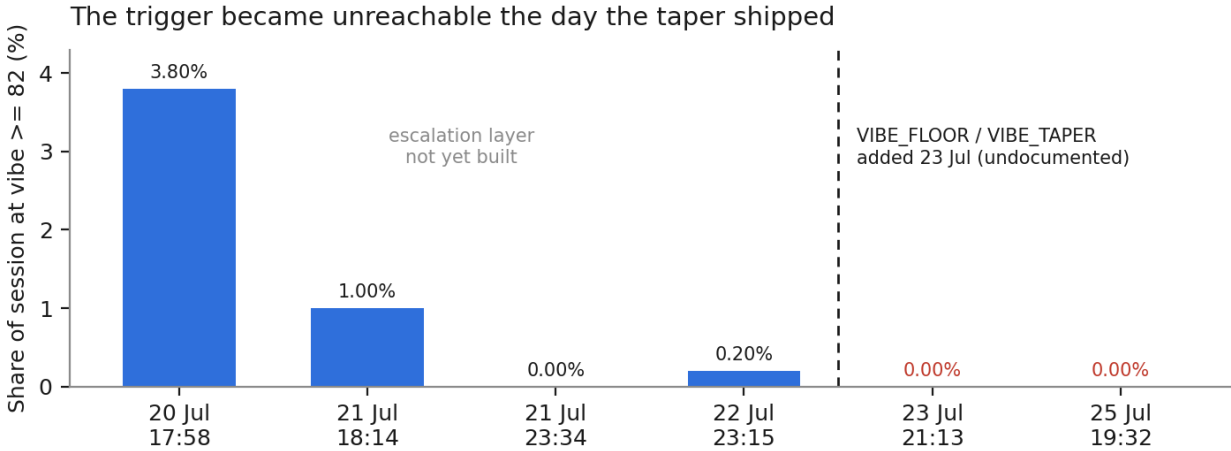
Two caveats on interpretation. First, these sessions are not a controlled experiment — they mix calibration runs, empty-room soak tests and live exhibition hours, and the report separates these where it matters. Second, hours 13:00–19:00 hold under one minute of data in total and are excluded from the hour-of-day figure.

Conformance against the spec

Spec reference	What the spec states	What the field data shows	Verdict
II, <i>Base activity</i>	“a single person crossing (~25% moving) reads ~0.4, so a maxed meter needs sustained, whole-frame motion”	97.2% of runtime below 30; 0.36% at or above 80	Behaves as written, but the resulting range is unusable

Spec reference	What the spec states	What the field data shows	Verdict
II, <i>Attack / release</i>	attack 1.6/s, release 0.5/s	A flat 11–16 baseline persists at every hour, including 03:00	Slow release over a noise floor keeps the meter permanently lit
II, <i>Per-section vibe</i>	each third gets an independent meter at gain 2.4	One third absorbed roughly 90% of scored energy on three nights	Mapping or gain is skewed, not the crowd
II, <i>Gesture and move bonuses</i>	heart +45, V +22, Y +9, shape bonus	DANCE_MOVE is the most frequent event in the corpus at 8,760	Conforms; the healthiest part of the pipeline
II, <i>Network / multi-venue sync</i>	sharedVibe = max(local, peers)	peers = 0 in all 17 sessions	Never exercised; unverified specification
II, <i>Outputs and logging</i>	1 Hz {shared, local, peers, l, m, r}	109,344 samples, schema matches, plus per-section scores	Conforms
—	(not in spec)	taperActivity(), peak/takeover mode, timed warning, creature cast	Four behaviours the spec does not describe

Finding 1 — an undocumented taper closed off the top of the scale



Red bars are the only two sessions in which the escalation layer was actually running.

Figure 1: Share of each session spent at or above the arming threshold of 82. The escalation layer only existed during the two red sessions, both of which sit at zero.

Part II documents a five-step pipeline in which `currentActivity()` feeds directly into the sensitivity scaling and the asymmetric EMA. The running implementation inserts an extra stage between the two:

```
t = (activity - VIBE_FLOOR) / (1 - VIBE_FLOOR)
activity' = t <= 0 ? 0 : min(1, t) ^ VIBE_TAPER
```

with `VIBE_FLOOR = 0.15` and `VIBE_TAPER = 1.2`. The intent was reasonable — sensor speckle was inflating the meter, and high vibe was meant to be harder to earn. The measured effect was categorical rather than gradual:

Period	Sessions	Samples	Time at vibe ≥ 82
Before 23 Jul (pre-taper)	4	61,475	0.56%
From 23 Jul (post-taper)	6	47,869	0.00%

Nothing about the venue changed between those two periods; the 22 July and 25 July sessions are both long evening runs on the same feeds. What changed was the transfer function. Combined with the escalation layer having been built during the same window, the result is that the trigger and its unreachability shipped together and the condition has never been observed in production.

Finding 2 — the scale is compressed into its bottom third

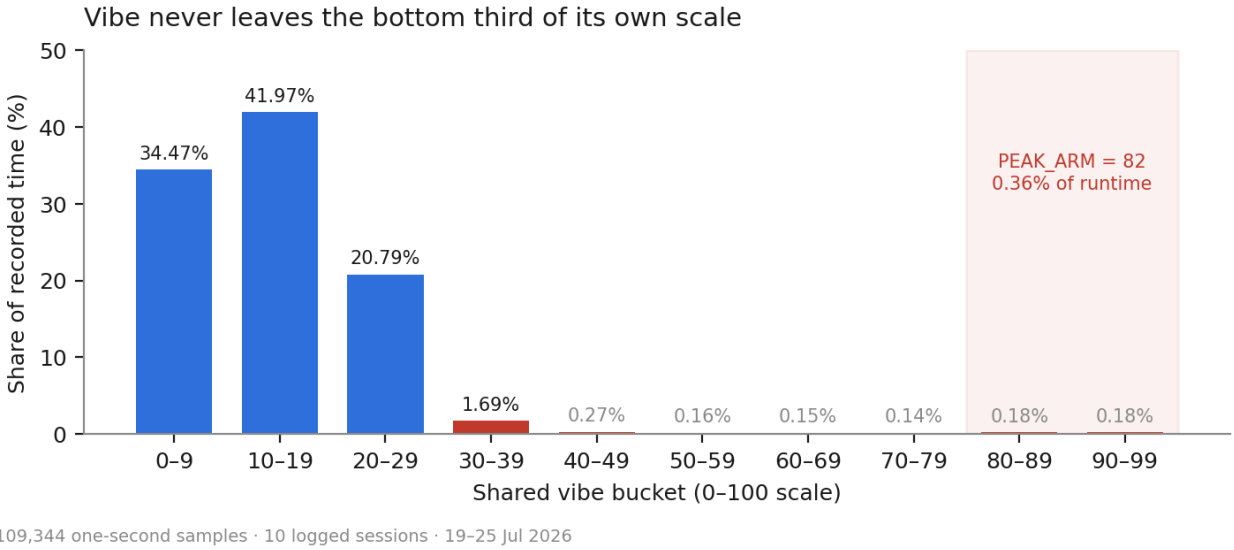


Figure 2: Distribution of shared vibe across all logged runtime. Blue bars mark the 0-29 band holding 97.2% of samples; the shaded region marks the arming threshold.

Even setting the taper aside, the activity function specified in Part II is

```
activity = blob * 1.5 + min(movers, 6) * 0.07
```

The spec is candid that this is “deliberately gentle”. The field data quantifies how gentle: a 25% frame-coverage crossing reads about 0.4, and the sustained whole-frame motion needed to approach 1.0 does not occur on a monitor feed of dancers. The consequences compound downstream — per-section creature stages past the midpoint rarely appear, and the upper half of a 0-100 scale carries no information.

Finding 3 — the baseline is noise, not people

A crowd metric should approach zero in an empty room. Instead the mean sits between 11 and 16 at nearly every hour of the cycle, including the small hours when the space is empty. That flatness is the signature of a detector integrating sensor speckle rather than bodies, and it is what the taper was introduced to suppress.

The taper did not eliminate it, which suggests VIBE_FLOOR is set roughly half as high as it needs to be. Only three narrow windows — midday, 20:00 and 23:00 — rise clearly above the noise band, and those are the genuine human peaks.

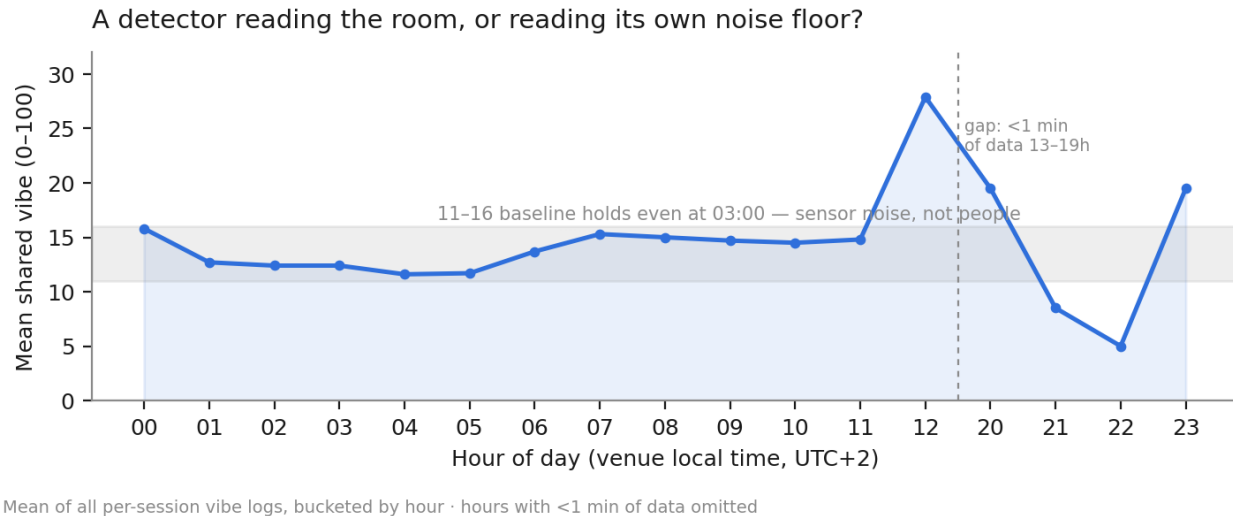


Figure 3: Mean shared vibe by hour of day. The shaded band marks the 11–16 range the meter holds around the clock.

Finding 4 — one section is eating the show

Part II describes three independent section meters feeding a per-section leaderboard. Accumulated vibe-seconds per session tell a different story:

Session	Left	Middle	Right	Dominant share
20 Jul 17:58	2,582	145	23,524	Right, 90%
21 Jul 18:14	1,235	5	14,345	Right, 92%
21 Jul 23:34	250	334	7	Middle, 57%
22 Jul 23:15	13,667	1,525	152,058	Right, 91%
23 Jul 21:13	625	139	21	Left, 80%
25 Jul 19:32	1,491	666	715	Left, 52%

A single third taking about 90% of the energy on three nights and near zero on another is not plausible crowd behaviour. The likely cause is the section-to-monitor mapping: when a section is bound to a specific monitor quad rather than a geometric third, all motion within that monitor is attributed to one section, and a busy screen monopolises the leaderboard. Until this is re-checked, per-section scores should not be read as a measurement of where the crowd actually is.

Finding 5 — what is working

Three parts of the spec are validated by the data and should not be touched:

- **The gesture layer.** DANCE_MOVE fired 8,760 times. The bonus mechanism described in Part II is the most reliably triggered path in the entire system, and it is what carried the piece while the vibe meter sat flat.
- **The logging contract.** The 1 Hz night log and the cross-day archive match the documented schema exactly, which is the only reason this report was possible.
- **Gateway resilience.** 2,097 successful reachability probes against 8 failures, and 30 successful feed connections against 12 handshake retries and 6 hard failures. The WHEP-then-WebSocket fallback is doing its job.

Finding 6 — the spec has drifted from the implementation

Four behaviours now exist in the running app that Part II does not describe. Each should be folded into the spec or explicitly declared out of scope:

1. **Activity tapering** — `taperActivity()`, the subject of Finding 1, sitting between steps 1 and 2 of the documented pipeline.
2. **Peak / takeover mode** — an arming threshold of 82 held for 1.2 s triggers a three-count and a 14-second forced-maximum window, re-arming only below 60. An entire event layer absent from the spec, and dead code in practice.
3. **A scheduled warning event** on a fixed 120-second timer, independent of vibe.
4. **The performer cast** — Part II states that per-section vibe “drives the trio of green-screen performers”. Those have been replaced by a three-stage creature cast whose stage is selected by section vibe.

Items 2 and 3 together produce the report’s most uncomfortable observation. The crowd-driven escalation fired **0** times; the clock-driven warning fired **152** times over the same period. Every dramatic beat the installation has produced so far came from a timer rather than from the room.

Recommended amendments

Ordered by impact on the exhibited work.

1. **Retune the taper before anything else.** `VIBE_FLOOR = 0.15` with `VIBE_TAPER = 1.2` currently removes the entire top of the range. Either lower the gamma to 1.0 and keep the floor, or keep the gamma and drop the arming threshold — but the two were tuned independently and their product was never checked against the trigger.
2. **Rescale the activity mapping, or lower the trigger.** Two options:
 - *Cheap*: set the arming threshold to 35–40. On the measured distribution that makes the event reachable a few times per busy hour instead of never.
 - *Correct*: rescale `currentActivity()` so a genuinely busy room maps to 70–100. Raising the blob coefficient from 1.5 toward roughly 3.0 would place observed peaks near the top of the scale while leaving the empty-room baseline where it is.
3. **Raise the noise floor to about 0.25–0.30** so an empty room reports near zero, which is what a crowd metric should do — and document the value in Part II either way.
4. **Audit the section-to-monitor mapping** before the per-section leaderboard is used for anything, and consider normalising each section by its own area.
5. **Document the taper and the escalation layer** in Part II so the spec again matches the implementation, and record calibration provenance for the new constants the way Parts I and II already do for the originals.
6. **Either exercise or descope multi-venue sync.** The `sharedVibe = max(local, peers)` path has never run with a peer present across 49 hours.

Appendix A — session detail

All ten sessions that produced vibe telemetry. “% dead” is the share of samples below vibe 5. Seven further short runs on 19–20 July predate vibe logging and contributed 89 snapshots and 94 subject IDs during setup.

Started	Min	Samples	Mean	Peak	>=50	>=82	Dead	Snaps	Faces	FPS
20 Jul 17:58	134.1	6,138	15.8	100	6.80	3.80	27.7	386	192	56
21 Jul 18:14	63.2	3,617	12.8	98	2.10	1.00	25.4	469	0	24
21 Jul 23:34	222.6	12,945	12.8	38	0.00	0.00	0.2	2,456	124	78
22 Jul 23:15	679.4	38,775	16.1	100	0.50	0.20	2.9	7,281	1,231	31
23 Jul 10:34	13.3	25	13.7	31	0.00	0.00	12.0	106	98	177
23 Jul 10:48	0.4	19	3.8	28	0.00	0.00	89.5	120	114	155
23 Jul 20:53	2.3	59	7.7	27	0.00	0.00	39.0	171	118	147
23 Jul 21:13	0.4	19	0.6	5	0.00	0.00	94.7	3	2	220
23 Jul 21:13	1,302.8	16,945	17.8	61	0.00	0.00	0.5	2,841	314	113
25 Jul 19:32	531.5	30,802	7.1	100	0.70	0.00	60.4	8,879	822	77

Columns: **Min** is session length in minutes, **>=50** / **>=82** / **Dead** are the share of samples above 50, above 82 and below 5 respectively. **Snaps** and **Faces** count **files**, which overstates distinct captures — see Appendix C.

Cross-day archive coverage, written by the retention job:

Day	Samples	Minutes	Mean	Peak
21 Jul	10,775	179.6	13.7	98
22 Jul	31,843	530.7	9.9	53
23 Jul	18,376	306.3	16.3	39

Appendix B — event census

Counts are of **log events**, which is not the same as the file counts used elsewhere in this report. **CLEAN_FACE** fired 704 times as an event, while 3,019 files carry the clean-face marker — the event is emitted once per qualifying detection rather than once per written file, and each detection can write two files (see Appendix C).

Event	Count	Meaning
DANCE_MOVE	8,760	Gesture or dance move scored
DIAG	7,813	Detector diagnostic sample
NEW_SUBJECT	6,139	New tracked subject ID
SNAPSHOT	5,116	Frame written to disk
GATEWAY_OK	2,097	go2rtc gateway reachable
CLEAN_FACE	704	Frontal, level, confident face
BLOOD_MOON + CAST_WARNING	91 + 61	Scheduled 120-second warning, before and after rename
MONITOR_SET	68	Monitor quad placed or moved
FEED_CONNECT	30	Sentinel / Sparkle stream connected
FEED_TRY / FEED_TRY_FAIL	18 / 12	Transport attempt and handshake retry
MONITOR_ADDED	15	New monitor region defined
SESSION_START	12	Session began
GATEWAY_FAIL	8	Gateway unreachable
CYCLE_MODE	8	Cycle mode toggled

Event	Count	Meaning
FEED_ERROR	6	Feed gave up after both transports
PEAK_MODE	0	Crowd-driven escalation event

Appendix C — storage and the double-write

Capture ran at 463 files per hour, averaging roughly 0.9 MB each — about 9 GB per 24 hours of runtime. The archive stands at 21 GB.

The raw file count overstates what was actually captured. When a detection is annotated, the unit writes both a marked and an unmarked copy of the same frame, so 3,093 captures occupy two files each:

	Files	Distinct captures
All snapshots	22,801	19,708
Carrying the clean-face marker	3,019	1,512
Share with a clean face	13.2%	7.7%

Of the 1,512 distinct face captures, 1,507 exist in both marked and unmarked form and 5 are unmarked only. The 7.7% figure is the honest one: **fewer than one frame in twelve contains a usable face**, and the apparent 13.2% yield is an artefact of counting files.

Two reductions follow. Deduplicating the paired writes removes 3,093 files at no information cost, and face-gated retention on top of that would discard the 92.3% of distinct captures that contain nothing usable.